ANALYTICAL REFLECTION

Analytical reflection: flexibility is not fuel security

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919 words, excluding title and references.

Can an airline improve energy security by buying fuel somewhere else? Fuel tankering means carrying additional fuel from an origin airport to reduce purchases later. The same action can respond to price differences or an interruption in supply. This project asks what destination price premium compensates the additional fuel consumed in transport, and how much destination refuelling demand a given extra uplift can displace. Its contribution is a reproducible threshold model linked to an East Asian policy comparison. It evaluates conditional trade-offs rather than claiming to identify profitable trades from unavailable airport prices.

The question matters because Asian aviation connects globally traded oil to infrastructure that must deliver a specific product at a specific time. EIA estimates that Asia received 89% of crude oil and condensate passing through Hormuz in the first half of 2025; this is not a measurement of jet fuel deliveries. The IEA's 2026 Southeast Asia outlook also identifies dependence on Middle Eastern crude and constraints on substituting refinery feedstocks. These sources establish exposure, not the probability that Changi, Hong Kong or Narita will run short. A national oil inventory cannot automatically become certified jet fuel at an airport hydrant. [4, 5]

The model compares buying one tonne at the destination with loading enough additional fuel at the origin to deliver that same tonne. Route, payload, reserves, onward requirements and terminal fuel position are held constant. ICAO reports an aircraft-dependent additional fuel penalty of 2.5–4.5% of added weight per flight hour. I use that range as a sensitivity parameter, with 3.5% as an illustrative midpoint. A linear retention rule is the main approximation; an exponential rule tests structural sensitivity. Neither is an ICAO-prescribed equation or an aircraft performance calibration. Combustion emissions use ICAO's factor of 3.16 tonnes of CO_2 per tonne of Jet A fuel. [1, 2]

At four hours, the central linear case retains 86% of extra origin fuel. Delivering one useful tonne therefore requires 1.163 tonnes of extra uplift, burns 0.163 tonnes and emits 0.514 tonnes of additional direct CO_2. With both purchase options available, and no additional fees or carbon charge, the destination price must exceed the origin price by 16.3% to break even. An illustrative USD 100 per tonne CO_2 shadow price raises that threshold to 21.4% when origin fuel is assumed to cost USD 1,000 per tonne. These currency values are scenarios, not observed prices. The exponential alternative reduces the no-carbon threshold to 15.0%, showing that model structure matters.

Three routes connect the mechanism to regional geography: Singapore–Hong Kong, Hong Kong–Narita and Singapore–Narita. Public airport coordinates determine great-circle distance;

assumed routing and groundspeed convert distance into an airborne-time proxy, not a timetable. Central no-carbon thresholds are 13.4%, 15.8% and 32.6%, respectively. Varying burn coefficients, routing and speed produces linear-model envelopes of 7.7–21.5%, 9.0–25.7% and 17.5–58.7%. These are finite scenario ranges, not confidence intervals. The comparison suggests that transporting an extra tonne becomes more costly over longer missions under these assumptions; it does not establish any route's actual profitability. [3]

A shortage changes the decision problem. With one tonne of extra origin headroom, the four-hour case displaces only 0.86 tonnes of destination uplift, burning 0.14 tonnes and emitting 0.442 tonnes of CO_2. This uses a different denominator from delivering one full tonne. The result measures physical substitution, not flights saved. Its value depends on usable origin supply, aircraft weight limits, payload trade-offs and the destination's remaining requirements. Under a shared fuel quota, carrying fuel consumes more total fuel than supplying the same useful quantity locally. Nevertheless, moving fuel toward a bottleneck could preserve service; the model cannot estimate that benefit without allocation and operational data.

Institutional differences help explain why flexibility at one airline does not equal security for a region. Hong Kong's fuel operator documents airport storage, hydrants and an emergency marine backup: installed capacity is evidence of infrastructure, not actual stock or days of cover. Singapore's announced SAF procurement criteria include supply-chain resilience and delivery certainty, illustrating a coordination mechanism rather than proving current supply adequacy. Japan's participation in the IEA's March 2026 stock-release action illustrates national emergency capacity, but those commitments were not airport-specific jet fuel inventories. The relevant policy chain therefore runs from crude and refining through product logistics to aircraft access. [6, 7, 9]

Policy should distinguish economic tankering from documented operational necessity. Europe's ReFuelEU rule provides a comparator: the 90% uplift requirement applies annually by operator and Union airport, with provisions for justified shortages; it is not an Asian rule or a per-flight restriction. If Asian authorities adopt anti-tankering or emergency-allocation rules, they should document justified operational exceptions without imposing new prior dispatch approval or compromising flight safety. Improving stock visibility, supplier coordination and delivery redundancy remains essential. Airlines should compare verified into-plane quotes only after engineering feasibility checks. Authorities should assess distributional effects too: shifting demand toward an origin may help one carrier while tightening supply for others. This project does not quantify that externality. [10, 11]

The principal limitation is missing transaction and operational evidence. Wholesale benchmarks cannot identify airport contract prices, taxes, credit terms or handling charges. Public observations are separated from literature parameters and analyst scenarios throughout the exhibit; no missing quote is silently imputed. A stronger study would combine dated supplier quotes, aircraft performance and weight data, airport inventories and disruption records, then test predictions against actual uplift decisions. For now, the defensible finding is narrower: tankering offers conditional spatial flexibility at a measurable fuel and emissions cost. Making that boundary visible is part of rigorous analysis, not a reason to disguise uncertainty. [12]

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